

Stormwater Regulation Summary: New York City, NY

Municipal Stormwater Management Technical Profile | Sempergreen USA

1. Regulatory Overview

- Primary driver: Combined Sewer Overflow (CSO) reduction — ~60% of NYC is served by combined sewers.
- Unified Stormwater Rule (USWR), effective February 2022, consolidated NYC DEP stormwater requirements.
- Applies to projects with $\geq 20,000$ SF soil disturbance OR $\geq 5,000$ SF new impervious surface.
- MS4 areas require Water Quality Volume (WQv) retention; CSS areas require Sewer Operations Volume control.
- Local Law 92/94 (2019) mandates sustainable roofing (solar, green roof, or combo) on all new buildings and major roof renovations.

2. Typical Soil Conditions

- Urban fill dominates most of Manhattan, Brooklyn, and Queens — highly variable composition.
- Native soils where present are primarily glacial till and outwash: sandy loam to silty clay.
- Hydrologic Soil Group: Predominantly C and D (poor infiltration) due to fill and compacted urban soils.
- Implication: On-site infiltration is rarely feasible — regulations emphasize retention and controlled release rather than ground infiltration.
- *Site-specific geotechnical investigation always required; fill composition varies block to block.*

3. Green Roof Requirements

- Green roofs are effectively mandatory under Local Law 92/94: all new buildings and major roof renovations must install sustainable roofing (green roof, solar, or combination) on 100% of the Sustainable Roofing Zone.
- NYC DEP Green Infrastructure Grant Program can reimburse up to 100% of green roof installation costs on qualifying sites.
- Green roofs receive stormwater retention credit under the USWR — credited volume depends on media depth and system design.
- Extensive green roofs (4" media) credited at approximately 1.0" retention; intensive systems receive higher credits.
- Bio-solar (PV + green roof) satisfies both LL 92/94 sustainable roofing and DEP stormwater requirements simultaneously.

4. TSS Requirements

- NYC DEP requires 80% Total Suspended Solids (TSS) removal for post-construction stormwater management.
- Green roofs are an approved BMP that receives TSS removal credit — typically credited at 80-85% TSS reduction.
- The USWR accepts green roofs as a standalone BMP for TSS compliance when properly designed.
- Additional TSS measures (hydrodynamic separators, media filters) may be required for runoff from non-green-roof impervious areas.
- CSS areas: TSS reduction contributes to CSO pollutant load reduction targets.

5. Nature-Based Retention Requirements

- MS4 areas: Retain 1.5" WQv (Water Quality Volume) from all impervious surfaces.
- CSS areas: Retain 1.85" Sewer Operations Volume (SOV) to reduce combined sewer discharge.
- Retention must be achieved through approved on-site BMPs — green roofs, bioretention, permeable pavement, or cisterns.
- Curve Number (CN) method is the standard evaluation metric: pre-development CN compared to post-development with BMPs.
- Target CN varies by land use; typical post-development target with green roof: CN 55-65 (approximating meadow/woods in good condition).
- *NYC uses a modified CN approach specific to the USWR — verify against current DEP technical guidance.*

6. Allowable Outflow Rates

- CSS areas: Maximum release rate = the greater of 0.25 cfs or 10% of Allowable Flow (if Allowable Flow < 0.25 cfs, release = Allowable Flow).
- MS4 areas: Post-development peak flow must not exceed the site's Allowable Flow for 1-yr, 10-yr, and 100-yr storms.
- Converted to metric: 0.25 cfs/acre $\approx$ 17.4 l/s/ha (typical small-site CSS limit).
- *Actual allowable rate is site-specific, calculated based on downstream sewer capacity — not a flat rate.*
- NYC does not use a simple l/s/ha standard; allowable flow is determined by DEP sewer capacity analysis for each connection point.

Items in *italic* should be verified against current municipal codes. | Generated by Sempergreen USA Stormwater BMP Comparison Tool